VIT CHENNAI
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Assignment -1

Database Systems Lab

BCSE302P

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Topic:

DML Commands

1. **STUDENT STUDENT (STUD_ID, STUD_NAME, GENDER, DEPT, YEAR, AGE, CGPA, CITY) Constraints: • STUD_ID → PRIMARY KEY • STUD_NAME → NOT NULL • GENDER → CHECK (GENDER IN ('M','F')) • YEAR → CHECK (YEAR BETWEEN 1 AND 4) • AGE → CHECK (AGE > 18) • CITY → DEFAULT 'CHENNAI'**

```
SQL> connect system
Enter password:
Connected.
SQL> CREATE TABLE STUDENT_24BYB1117 (
  2      STUD_ID INT PRIMARY KEY,
  3      STUD_NAME VARCHAR(50) NOT NULL,
  4      GENDER CHAR(1) CHECK (GENDER IN ('M','F')),
  5      DEPT VARCHAR(20),
  6      YEAR INT CHECK (YEAR BETWEEN 1 AND 4),
  7      AGE INT CHECK (AGE > 18),
  8      CGPA DECIMAL(3,2),
  9      CITY VARCHAR(30) DEFAULT 'CHENNAI'
 10  );

Table created.
```

2. **COURSE COURSE (COURSE_ID, COURSE_NAME, CATEGORY, DURATION, CREDITS, FEES) Constraints: • COURSE_ID → PRIMARY KEY • COURSE_NAME → NOT NULL • CATEGORY → CHECK (CATEGORY IN ('UG','PG')) • DURATION → CHECK (DURATION >= 1) • CREDITS → CHECK (CREDITS > 0)**

```
SQL> CREATE TABLE COURSE_24BYB1117 (
  2      COURSE_ID INT PRIMARY KEY,
  3      COURSE_NAME VARCHAR(50) NOT NULL,
  4      CATEGORY VARCHAR(2) CHECK (CATEGORY IN ('UG','PG')),
  5      DURATION INT CHECK (DURATION ≥ 1),
  6      CREDITS INT CHECK (CREDITS > 0),
  7      FEES DECIMAL(10,2)
  8  );

Table created.
```

3. **FACULTY FACULTY (FAC_ID, FAC_NAME, GENDER, QUALIFICATION, EXPERIENCE, SALARY, CITY) Constraints: • FAC_ID → PRIMARY KEY • FAC_NAME → NOT NULL • GENDER → CHECK (GENDER IN ('M','F')) • EXPERIENCE → CHECK (EXPERIENCE >= 0) • CITY → DEFAULT 'CHENNAI'**

```
SQL> CREATE TABLE FACULTY_24BYB1117 (
  2      FAC_ID INT PRIMARY KEY,
  3      FAC_NAME VARCHAR(50) NOT NULL,
  4      GENDER CHAR(1) CHECK (GENDER IN ('M','F')),
  5      QUALIFICATION VARCHAR(30),
  6      EXPERIENCE INT CHECK (EXPERIENCE ≥ 0),
  7      SALARY DECIMAL(10,2),
  8      CITY VARCHAR(30) DEFAULT 'CHENNAI'
  9  );

Table created.
```

A. INSERT OPERATIONS

1. Insert a student record specifying all column values.

```
SQL> INSERT INTO STUDENT_24BYB1117
  2  VALUES (101,'Rakshit','M','CSE',2,20,9.5,'CHENNAI');
```

2. Insert a student record without specifying CITY (DEFAULT should apply).

```
SQL> INSERT INTO STUDENT_24BYB1117
  2  (STUD_ID, STUD_NAME, GENDER, DEPT, YEAR, AGE, CGPA)
  3  VALUES (102,'ANITA','F','ECE',3,21,9.1);
```

3.Insert three student records using a single INSERT statement.

```
SQL> INSERT ALL
  2    INTO STUDENT_24BYB1117 VALUES (103,'ARJUN','M','MECH',1,19,7.8,'CHENNAI')
  3    INTO STUDENT_24BYB1117 VALUES (104,'SNEHA','F','CSE',4,22,8.9,'HYDERABAD')
  4    INTO STUDENT_24BYB1117 VALUES (105,'VIKRAM','M','EEE',2,20,8.0,'CHENNAI')
  5  SELECT * FROM dual;

3 rows created.
```

4. Insert a course record using column list in a different order.

```
SQL> INSERT INTO COURSE_24BYB1117
  2  (COURSE_NAME, COURSE_ID, CATEGORY, DURATION, CREDITS, FEES)
  3  VALUES ('DBMS',201,'UG',1,4,30000);

1 row created.
```

5. Insert multiple course records in a single statement.

```
SQL> INSERT ALL
  2    INTO COURSE_24BYB1117 VALUES (202,'AI','PG',2,6,90000)
  3    INTO COURSE_24BYB1117 VALUES (203,'ML','PG',2,5,85000)
  4  SELECT * FROM dual;

2 rows created.
```

6. Insert a faculty record specifying all attributes.

```
SQL> INSERT INTO FACULTY_24BYB1117
  2  VALUES (301,'DR KUMAR','M','PhD',10,120000,'CHENNAI');

1 row created.
```

7. Insert a faculty record without specifying CITY.

```
SQL> INSERT INTO FACULTY_24BYB1117
  2  (FAC_ID, FAC_NAME, GENDER, QUALIFICATION, EXPERIENCE, SALARY)
  3  VALUES (302,'DR MEENA','F','PhD',8,110000);

1 row created.
```

8. Insert a faculty record using substitution variables.

```
SQL> INSERT INTO FACULTY_24BYB1117
  2  VALUES (&FAC_ID,'&FAC_NAME','&GENDER','&QUALIFICATION',&EXPERIENCE,&SALARY,'&CITY');
Enter value for fac_id: 303
Enter value for fac_name: Tahir
Enter value for gender: M
Enter value for qualification: PhD
Enter value for experience: 10
Enter value for salary: 150000
Enter value for city: CHENNAI
old   2: VALUES (&FAC_ID,'&FAC_NAME','&GENDER','&QUALIFICATION',&EXPERIENCE,&SALARY,'&CITY')
new   2: VALUES (303,'Tahir','M','PhD',10,150000,'CHENNAI')

1 row created.
```

9. Insert records into all three tables ensuring all CHECK constraints are satisfied.

```
SQL> INSERT INTO STUDENT_24BYB1117
  2  VALUES (201,'RAKSHIT','M','CSE',2,20,9.6,'CHENNAI');

1 row created.

SQL> INSERT INTO COURSE_24BYB1117
  2  VALUES (301,'DATABASE SYSTEMS','UG',1,4,35000);

1 row created.
```

```
SQL> INSERT INTO FACULTY_24BYB1117
  2    VALUES (501,'DR GEETHA','F','PhD',15,150000,'CHENNAI');

1 row created.
```

10. Attempt to insert a student with AGE less than or equal to 18 and observe the error.

```
SQL> INSERT INTO STUDENT_24BYB1117
  2  VALUES (107,'MINOR','M','CSE',1,18,7.0,'CHENNAI');
INSERT INTO STUDENT_24BYB1117
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.SYS_C007000) violated
```

B. UPDATE OPERATIONS

1. Update CGPA of a student using STUD_ID.

```
SQL> UPDATE STUDENT_24BYB1117
  2  SET CGPA = 9.6
  3  WHERE STUD_ID = 101;

1 row updated.
```

2. Update CITY of all students belonging to a specific department.

```
SQL> UPDATE STUDENT_24BYB1117
  2  SET CITY = 'BANGALORE'
  3  WHERE DEPT = 'CSE';

3 rows updated.
```

3. Increment YEAR by 1 for all final-year students.

```
SQL> UPDATE STUDENT_24BYB1117
  2  SET YEAR = YEAR + 1
  3  WHERE YEAR = 4;
UPDATE STUDENT_24BYB1117
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.SYS_C006999) violated
```

4. Update COURSE fees by increasing 10% for PG courses.

```
SQL> UPDATE COURSE_24BYB1117
  2  SET FEES = FEES * 1.10
  3  WHERE CATEGORY = 'PG';

2 rows updated.
```

5. Update COURSE duration for a given COURSE_ID.

```
SQL> UPDATE COURSE_24BYB1117
  2  SET DURATION = 3
  3  WHERE COURSE_ID = 202;

1 row updated.
```

6. Update faculty salary based on EXPERIENCE.

```
SQL> UPDATE FACULTY_24BYB1117
  2  SET SALARY = SALARY + 20000
  3  WHERE EXPERIENCE > 5;

3 rows updated.
```

7. Update CITY to 'BANGALORE' for female faculty members.

```
SQL> UPDATE FACULTY_24BYB1117
  2  SET CITY = 'BANGALORE'
  3  WHERE GENDER = 'F';

1 row updated.
```

8. Update multiple columns (SALARY and CITY) for a single faculty record.

```
SQL> UPDATE FACULTY_24BYB1117
  2  SET SALARY = 130000,
  3      CITY = 'MYSORE'
  4  WHERE FAC_ID = 301;

1 row updated.
```

9. Update student names to uppercase.

```
SQL> UPDATE STUDENT_24BYB1117
  2  SET STUD_NAME = UPPER(STUD_NAME);

6 rows updated.
```

10. Attempt to update COURSE category to an invalid value and note the error.

```
SQL> UPDATE COURSE_24BYB1117
  2  SET CATEGORY = 'DIPLOMA'
  3  WHERE COURSE_ID = 202;
SET CATEGORY = 'DIPLOMA'
              *
ERROR at line 2:
ORA-12899: value too large for column "SYSTEM"."COURSE_24BYB1117"."CATEGORY" (actual: 7, maximum: 2)
```

After Updation

```
SQL> SELECT * FROM STUDENT_24BYB1117;

STUD_ID STUD_NAME       GENDER DEPT        YEAR  AGE  CGPA CITY
------- --------------- ------ ---------- ----- ---- ----- ---------------
    101 RAKSHIT         M      CSE            3   20  9.60 BANGALORE
    102 ANITA           F      ECE            4   21  9.10 CHENNAI
    103 ARJUN           M      MECH           2   19  7.80 CHENNAI
    104 SNEHA           F      CSE            4   22  8.90 BANGALORE
    105 VIKRAM          M      EEE            3   20  8.00 CHENNAI
    106 RAVI            M      CSE            4   21  8.20 BANGALORE

6 rows selected.

SQL> SELECT * FROM COURSE_24BYB1117;

 COURSE_ID COURSE_NAME                                          CA   DURATION    CREDITS       FEES
---------- ------------------------------------------------- -- ---------- ---------- ----------
       201 DBMS                                              UG          1          4      30000
       202 AI                                                PG          3          6      99000
       203 ML                                                PG          2          5      93500

SQL> SELECT * FROM FACULTY_24BYB1117;

    FAC_ID FAC_NAME                                          GENDER QUALIFICATION                  EX
PERIENCE           SALARY CITY
---------- ------------------------------------------------- ------ ------------------------------ --
-------- ---------- ---------------
       301 DR KUMAR                                          M      PhD
   10      130000 MYSORE
       302 DR MEENA                                          F      PhD
    8      130000 BANGALORE
       303 Tahir                                             M      PhD
   10      170000 CHENNAI
```

C. DELETE OPERATIONS

1. Delete a student record using STUD_ID.

```
SQL> DELETE FROM STUDENT_24BYB1117
  2  WHERE STUD_ID = 101;

1 row deleted.
```

2. Delete all students belonging to YEAR = 1.

```
SQL> DELETE FROM STUDENT_24BYB1117
  2  WHERE YEAR = 1;

0 rows deleted.
```

3. Delete students whose CGPA is below 5.0.

```
SQL> DELETE FROM STUDENT_24BYB1117
  2  WHERE CGPA < 5.0;

0 rows deleted.
```

4. Delete a course using COURSE_ID.

```
SQL> DELETE FROM COURSE_24BYB1117
  2  WHERE COURSE_ID = 201;

1 row deleted.
```

5. Delete all courses with duration less than 2 years.

```
SQL> DELETE FROM COURSE_24BYB1117
  2  WHERE DURATION < 2;

0 rows deleted.
```

6. Delete faculty members with EXPERIENCE less than 1 year.

```
SQL> DELETE FROM FACULTY_24BYB1117
  2  WHERE EXPERIENCE < 1;

0 rows deleted.
```

7. Delete faculty belonging to a specific CITY.

```
SQL> DELETE FROM FACULTY_24BYB1117
  2  WHERE CITY = 'BANGALORE';

1 row deleted.
```

8. Delete all student records from the table.

```
SQL> DELETE FROM STUDENT_24BYB1117;

5 rows deleted.
```

9. Rollback the deletion (if transaction control is enabled).

```
SQL> ROLLBACK;

Rollback complete.
```

10. Attempt to delete a record that does not exist and observe the result.

```
SQL> DELETE FROM STUDENT_24BYB1117
  2  WHERE STUD_ID = 999;

0 rows deleted.
```

After deletion of complete table, a few more rows are added back for Select operations

```
SQL> INSERT ALL
  2    INTO STUDENT_24BYB1117 VALUES (101,'RAKSHIT','M','CSE',2,20,9.6,'CHENNAI')
  3    INTO STUDENT_24BYB1117 VALUES (102,'ARJUN','M','MECH',1,19,7.8,'CHENNAI')
  4    INTO STUDENT_24BYB1117 VALUES (103,'SNEHA','F','CSE',3,21,8.9,'HYDERABAD')
  5    INTO STUDENT_24BYB1117 VALUES (104,'VIKRAM','M','EEE',2,20,8.0,'CHENNAI')
  6    INTO STUDENT_24BYB1117 VALUES (105,'ANITA','F','ECE',4,22,9.1,'BANGALORE')
  7  SELECT * FROM dual;

5 rows created.
```

D. SELECT OPERATIONS

1. Display all student details.

```
SQL> SELECT * FROM STUDENT_24BYB1117;

STUD_ID STUD_NAME       GENDER   DEPT        YEAR  AGE  CGPA CITY
------- --------------- -------- ---------- ----- ---- ----- ---------------
    101 RAKSHIT         M        CSE            2   20  9.60 CHENNAI
    102 ARJUN           M        MECH           1   19  7.80 CHENNAI
    103 SNEHA           F        CSE            3   21  8.90 HYDERABAD
    104 VIKRAM          M        EEE            2   20  8.00 CHENNAI
    105 ANITA           F        ECE            4   22  9.10 BANGALORE
```

2. Display STUD_NAME, DEPT and CGPA of students.

```
SQL> SELECT STUD_NAME, DEPT, CGPA
  2  FROM STUDENT_24BYB1117;

STUD_NAME       DEPT         CGPA
--------------- ---------- -----
RAKSHIT         CSE          9.60
ARJUN           MECH         7.80
SNEHA           CSE          8.90
VIKRAM          EEE          8.00
ANITA           ECE          9.10
```

3. Display students whose AGE is greater than 20.

```
SQL> SELECT *
  2  FROM STUDENT_24BYB1117
  3  WHERE AGE > 20;

STUD_ID STUD_NAME       GENDER   DEPT        YEAR  AGE  CGPA CITY
------- --------------- -------- ---------- ----- ---- ----- ---------------
    103 SNEHA           F        CSE            3   21  8.90 HYDERABAD
    105 ANITA           F        ECE            4   22  9.10 BANGALORE
```

4. Display students ordered by CGPA in descending order.

```
SQL> SELECT *
  2  FROM STUDENT_24BYB1117
  3  ORDER BY CGPA DESC;

STUD_ID STUD_NAME       GENDER   DEPT        YEAR  AGE  CGPA CITY
------- --------------- -------- ---------- ----- ---- ----- ---------------
    101 RAKSHIT         M        CSE            2   20  9.60 CHENNAI
    105 ANITA           F        ECE            4   22  9.10 BANGALORE
    103 SNEHA           F        CSE            3   21  8.90 HYDERABAD
    104 VIKRAM          M        EEE            2   20  8.00 CHENNAI
    102 ARJUN           M        MECH           1   19  7.80 CHENNAI
```

5. Display distinct CITY values from STUDENT table.

```
SQL> SELECT DISTINCT CITY
  2  FROM STUDENT_24BYB1117;

CITY
----------------
CHENNAI
HYDERABAD
BANGALORE
```

6. Display all course details.

```
SQL> INSERT ALL
  2    INTO COURSE_24BYB1117 VALUES (204,'DATA SCIENCE','PG',2,6,95000)
  3    INTO COURSE_24BYB1117 VALUES (205,'CYBER SECURITY','PG',2,5,88000)
  4    INTO COURSE_24BYB1117 VALUES (206,'CLOUD COMPUTING','UG',1,4,42000)
  5    INTO COURSE_24BYB1117 VALUES (207,'IOT','UG',1,3,35000)
  6  SELECT * FROM dual;

4 rows created.

SQL> SELECT * FROM COURSE_24BYB1117;

COURSE_ID COURSE_NAME     CATEGORY DURATION CREDITS    FEES
--------- --------------- -------- -------- ------- -------
      204 DATA SCIENCE    PG              2       6   95000
      205 CYBER SECURITY  PG              2       5   88000
      206 CLOUD COMPUTING UG              1       4   42000
      207 IOT             UG              1       3   35000
```

7. Display courses belonging to PG category.

```
SQL> SELECT *
  2  FROM COURSE_24BYB1117
  3  WHERE CATEGORY = 'PG';

COURSE_ID COURSE_NAME     CATEGORY DURATION CREDITS    FEES
--------- --------------- -------- -------- ------- -------
      204 DATA SCIENCE    PG              2       6   95000
      205 CYBER SECURITY  PG              2       5   88000
```

8. Display faculty details ordered by EXPERIENCE.

```
SQL> INSERT ALL
  2    INTO FACULTY_24BYB1117 VALUES (401,'DR KUMAR','M','PhD',10,120000,'CHENNAI')
  3    INTO FACULTY_24BYB1117 VALUES (402,'DR MEENA','F','PhD',8,110000,'BANGALORE')
  4    INTO FACULTY_24BYB1117 VALUES (403,'DR RAVI','M','M.Tech',5,90000,'CHENNAI')
  5    INTO FACULTY_24BYB1117 VALUES (404,'DR SNEHA','F','PhD',12,140000,'HYDERABAD')
  6  SELECT * FROM dual;

4 rows created.

SQL> SELECT * FROM FACULTY_24BYB1117 ORDER BY EXPERIENCE;

FAC_ID FAC_NAME        GENDER   QUALIFICATIO EXPERIENCE  SALARY CITY
------ --------------- -------- ------------ ---------- ------- ---------------
   403 DR RAVI         M        M.Tech                5   90000 CHENNAI
   402 DR MEENA        F        PhD                   8  110000 BANGALORE
   401 DR KUMAR        M        PhD                  10  120000 CHENNAI
   404 DR SNEHA        F        PhD                  12  140000 HYDERABAD
```

9. Display FAC_NAME and SALARY of faculty working in CHENNAI.

```
SQL> SELECT FAC_NAME, SALARY
  2  FROM FACULTY_24BYB1117
  3  WHERE CITY = 'CHENNAI';

FAC_NAME          SALARY
---------------- -------
DR KUMAR          120000
DR RAVI            90000
```

10. Display faculty members with salary greater than a given amount.

```
SQL> SELECT *
  2  FROM FACULTY_24BYB1117
  3  WHERE SALARY > 120000;

FAC_ID FAC_NAME        GENDER   QUALIFICATIO EXPERIENCE  SALARY CITY
------ --------------- -------- ------------ ---------- ------- ---------------
   404 DR SNEHA        F        PhD                  12  140000 HYDERABAD
```